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..Anantha Vidyanikethana
Midterm examination - SEP - 2025
PHYSICS - Grade 10
(SCIENCE PAPER 1)

Maximum Marks: 80 **Time allowed: Two hours**
Date:19/9/2025

1. Answers to this Paper must be written on the paper provided separately.
2. You will not be allowed to write during the first 15 minutes.
3. This time is to be spent in reading the question paper.
4. The time given at the head of this Paper is the time allowed for writing the answers.
5. Section A is compulsory. Attempt any four questions from Section B.
6. The intended marks for questions or parts of questions are given in brackets [ ].

SECTION A (40 Marks)
(Attempt **all** questions from this Section.)

Question 1
Choose the correct answers to the questions from the given options. (Do not copy the question. Write the correct answers only.) **[15]**

i) A wheel is rotated around a fixed axis in an anticlockwise direction. If a tangential force is applied in the opposite direction, the moment of force acting on the wheel will be taken as

a) positive b) negative c) zero d) maximum

ii) A body of mass M kg placed at a height of 20 cm from the ground is shifted to 1.2 m vertically upwards. The increased potential energy will be,

a) 1M kgfm b) 1M gfcm c) 1M Nm d) 24M J

iii) During power production in a Nuclear-based thermoelectric power plant, the correct sequence of energy conversions taking place is:

a) Nuclear → mechanical → electrical
b) Chemical → heat → electrical
c) Heat → mechanical → electrical
d) Nuclear → heat → electrical

iv) Sushma, while using a block and tackle pulley system with a convenient direction to lift a load of 100 kgf using 20 kgf, she fixed 4 pulleys in the movable block. Which of the following statements is not correct for this situation?

a) M A < V R
b) The efficiency of the pulley system will decrease
c) It is convenient to exert effort.
d) She was unable to arrange the pulleys properly.

v) Some people are swimming in a pool. Which of the following phenomena is suitable for the appearance of legs in a poll?

a) Refraction of light
b) Total internal reflection of light
c) Reflection of light
d) Rectilinear propagation of light



vi) A convex lens of focal length 12 cm is placed between an object and a screen. If the image obtained on the screen is twice the length of the object, which of the following is suitable for the image formed

a) Image is beyond 24 cm
b) Image is beyond 12 cm
c) Image is at 24 cm
d) Image is between 12 cm and 24 cm

vii) Anitha is experimenting in a laboratory using a sound instrument. When it is placed in an enclosed box and played, it produces a loud sound. The reason is

a) The density of air in the box is less. c) The natural frequency of air matches the instrument
b) The amplitude of the instrument increases d) Both b and c

viii) In the Ohm's law experiment, the graph Voltage vs electric current is constant for a resistor. If the temperature of the resistor increases with time, then
a) The slope remains the same, but negative c) The slope is the tangent of the curve
b) The slope is zero d) The slope increases

ix) The heat capacity of a body depends on
a) Mass of the body c) change in temperature of the body
b) Both a and c d) density of the body

x) A nucleus of an atom consists of 146 neutrons and 95 protons. It decays after emitting an alpha particle. How many protons and neutrons are left in the nucleus after an alpha emission?
a) protons = 93, neutrons = 142 c) protons = 95, neutrons = 144
b) protons = 93, neutrons = 144 d) protons = 95, neutrons = 142

xi) In this picture, light rays are coming out of optical fibers. The phenomenon of light observed is
a) Refraction of light c) refraction of light
b) Total internal reflection d) dispersion of light



xii) For which of the following does the centre of gravity lie outside the object?
a) A pencil b) a dice c) a bangle d) a shotput

xiii) An electric current of 2 A is passed through a resistor of resistance 100 ohm.. The potential difference across the resistor is
a) 200 V b) 25 V c) 50 V d) 20 V

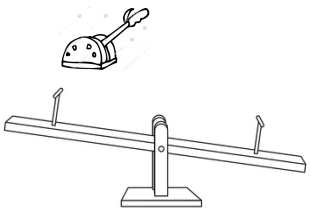
xiv) The focal length of a convex lens is 50 cm, then the power of the lens is
a) + 0.02 D b) - 0.02 D c) + 2 D d) - 2 D

xv) Which of the following radio isotopes is used in the testing of soil fertility
a) Cobalt-60 b) Potassium - 42 c) Iodine-131 d) Carbon-14

Question 2

i) Complete the following by choosing the correct answers from the brackets: [6]
(a) A nut cracker is an example of _____ [Class II / Class III] lever.
(b) The lens that produces an image bigger than the object placed in front of it is a [convex lens/concave lens]
(c) Unit of heat capacity is _____[NK^{-1} / NmK^{-1})]
(d) The threshold frequency of human hearing is _____ [100Hz / 20Hz]
(e) Two copper wires can have different resistivity if they have different _____. [lengths/temperatures]
(f) A radioactive radiation that affects both mass and atomic number of the parent element if it emits from the element is _____ [alpha/beta]

ii) Match the Levers in **Column A** to the class of lever in **Column B**. [2]



Class 1

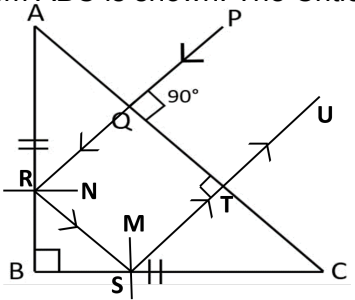
Class 2

Class 3'

(iii) a) Name the waves used in the SONAR device. [2]
b) Give a reason for using the waves above in the SONAR device.

Question 3

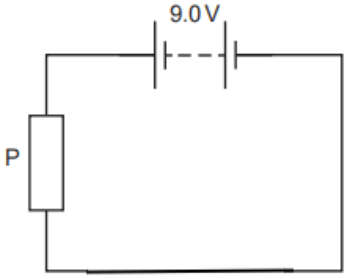
i) In this diagram, the path of a light ray P through a prism ABC is shown. The Critical angle of the prism is 42°.
a) Name the phenomenon of light in the prism [2]
b) What is the angle of incidence at R



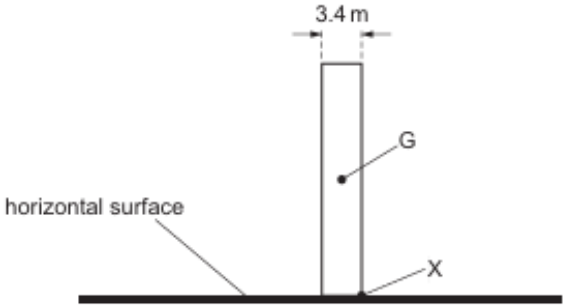
ii) The given figure shows that the circuit includes a battery of (e.m.f.) 9.0 V and a resistor P. The resistance P is 4.0 ohm. The current in P is 1.5 A. Calculate

a) The magnitude X of the charge flowing through P in 600 s. [2]

b) The potential difference at P.



iii) A large rectangular block of stone has a square base of side 3.4 m. The figure shows the block at rest on a horizontal surface. The block is of uniform density, and the centre of gravity G is at its centre. [2]



The weight of the block is 2500 N. Calculate the moment of the weight of the block about corner X.

iv) x g of water at 360 K is taken in a beaker. It is heated to its boiling point. In the process if 2730 J of heat is utilised then Find x. (Specific heat capacity of water is 4200 $Jkg^{-1}K^{-1}$, Assume there is no loss of energy)

v) A current flows through a metallic conductor for a long period of time. State the change you would expect in its:

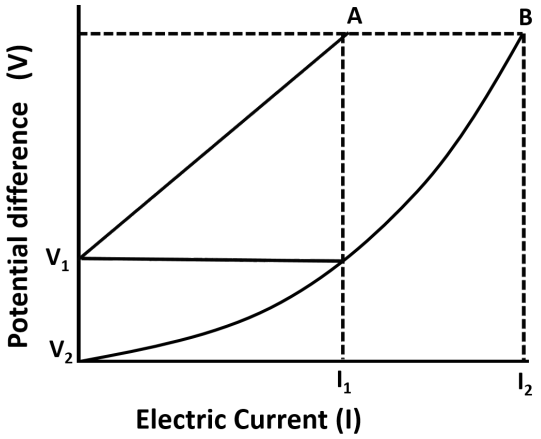
(a) Resistance

(b) Resistivity [2]

vi) Observe the V-I circuit diagram of two conductors, A and B. Answer the questions that follow. [2]

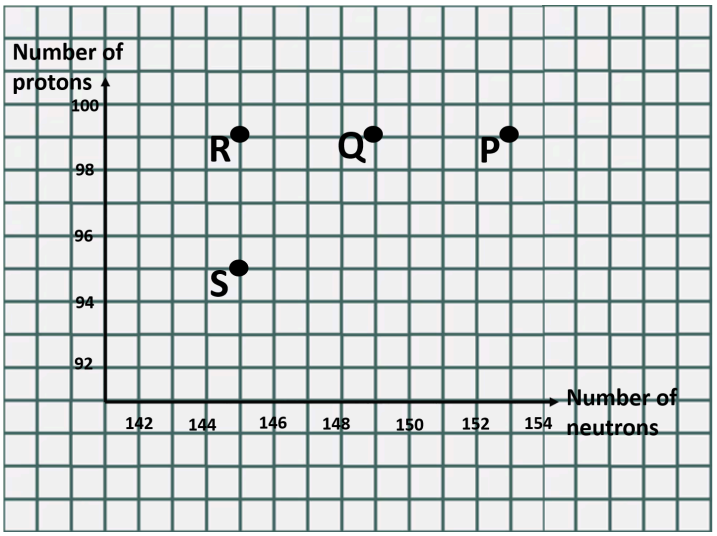
a) State the voltage at which the resistance of both A and B conductors is the same.

b) Which resistor has constant resistivity?



vii) Curium is a radioactive element with the symbol Es_{99}^{252} named in honour of Einstein. The graph of the

number of protons vs the number of neutrons for some elements is shown below: [3]



(a) Which point on the graph indicates the element Es?
(b) Which point on the graph indicates the daughter nucleus after Es undergoes radioactive decay of 2 α followed by 4 β?
(c) State the mass number of the daughter nucleus.

SECTION A (40 Marks)
(Attempt **any 4** questions from this Section.)

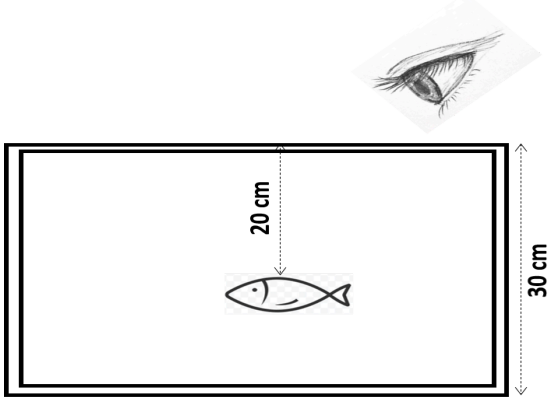
Question 4

i) In the diagram given below, the red and green lights are passing through an equilateral prism. The total internal reflection of red light is shown in the figure. [3]

a) Find the critical angle of the prism?
b) Copy the diagram to complete the path of the green light till it emerges from the prism.

(ii) A fish is swimming at a depth of 20 cm in an aquarium of depth 30 cm as shown in the figure.

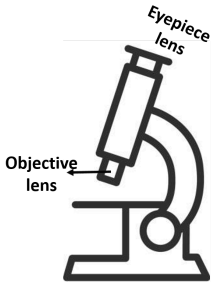
a) If the refractive index of water is 4/3, find the depth at which the fish appears from the outside of the aquarium. [3]
b) Find the shift of the fish that appears from the bottom of the aquarium.



iii) A lens is used in the compound microscope as the objective lens to produce a real, enlarged image. [4]

a) Name the lens used as the objective lens in this device.

b) Draw a ray diagram to show the image formed by the above-stated lens used in the device.



Question 5

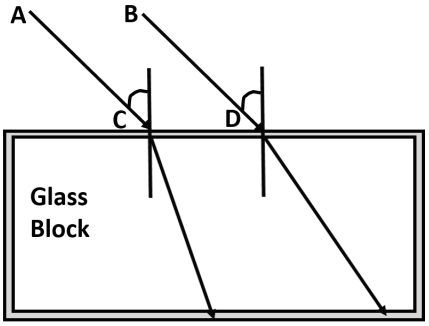
i) An object is placed in front of a concave lens at a distance of 45 cm from it. If its image is formed at a distance of 30 cm from the lens, calculate the focal length of the lens. [3]

ii) Two rays AC and BD are incident on a rectangular glass block as shown in the diagram. Observe the diagram and answer the questions that follow. [3]

a) Which of these two rays will have a greater wavelength?

b) Which of these two rays will have greater lateral displacement?

c) Can they travel parallel to each other, on emerging from the block?



iii) a) Name the radiations: [4]

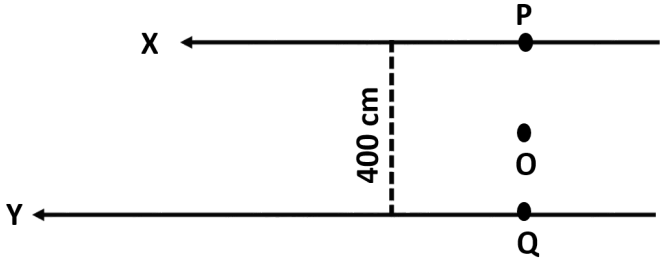
1. Used to take photographs in fog.
2. For that, a rock salt prism is used to study the spectrum.
3. Used to verify real and artificial diamonds.

b) Give one property common to all electromagnetic radiations.

Question 6

i) X and Y are two forces 2 N and 3 N respectively, 400 cm apart. O is the point 250 cm away from point P on X as shown in the figure. Calculate the sum of moments of X and Y about:

a) O b) P [3]



ii) By observing the figure shown here, answer the following questions [3]

a) Identify the liver used by the person to lift a load

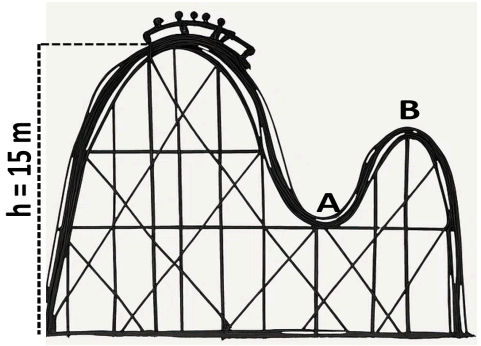
b) State the velocity ratio of the above-stated lever.

c) How do you increase the Mechanical advantage of the lever given in the figure?



iii) A train of mass 200 kg is moving down from a roller coaster from a height of 15 m above the ground, as shown in the figure. It reaches A at a speed of 10 ms^{-1}. Calculate: [4]

a) Kinetic energy of the train before leaving A
b) Vertical height of B from ground (take g = 10 ms^{-2} and neglect friction)
c) Kinetic energy of the train before reaching the ground (if 1000 J of energy is lost due to friction from B to the ground)



Question 7

i) Draw a block and tackle pulley of VR = 5 with a convenient direction of effort. [3]

ii) A person is going to find his mobile in the home that is misplaced. He asked his wife to give a call to his number. So he identifies the mobile when it starts ringing. [3]

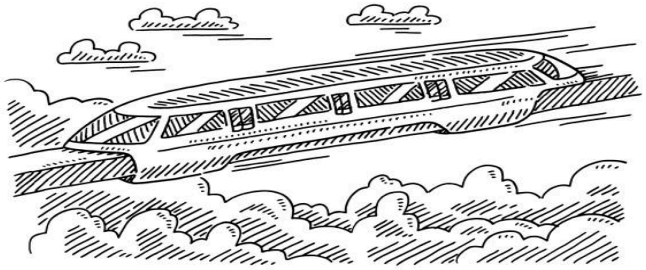
a) Name the characteristic of sound helpful in finding his mobile.
b) If the amplitude of the above characteristic is increased to three times, compare their loudness.
c) State one phenomenon that helps to increase the above characteristic without changing the pitch.

iii) A submarine in the sea sends an ultrasonic ping, and a stopwatch is started. simultaneously. The stopwatch stops on receiving the reflected wave from an obstacle and reads 1 minute 40 seconds. Calculate: [4]

a) The distance of the obstacle from the submarine (Speed of sound in water 1500 ms−1).
b) The time taken to receive the echo from another obstacle at the same distance calculated in question (a), on the land, from a source of a sound (Speed of sound in air is 320 ms^{-1}).

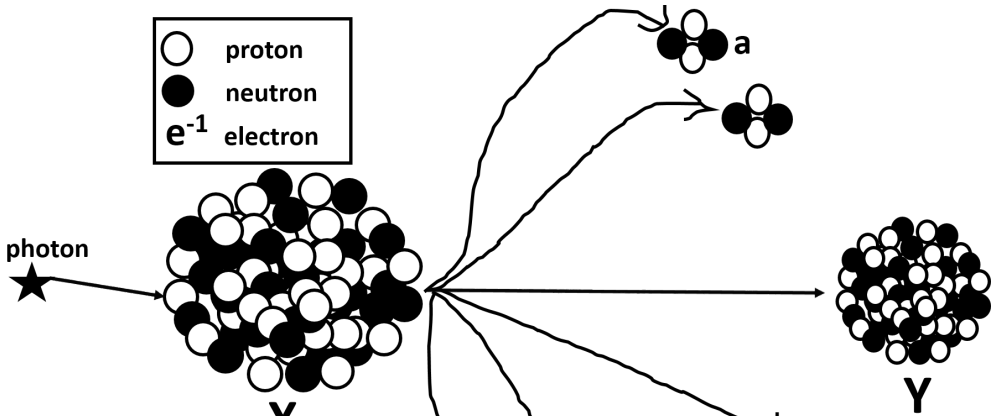
Question 8

i) In this picture, you can see a maglev train, which ran between Tokyo and Osaka in 1986 at a speed of 320 km/h. It was moving 10 cm above the track level. [3]



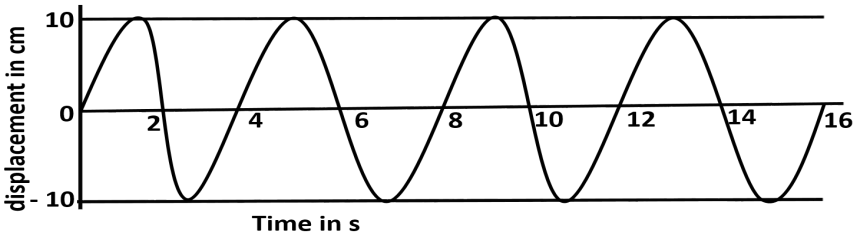
a) Name and explain the conductor used in this train for its magnetic leviation.
b) Can it be available in normal temperatures?

ii) Consider the following reaction and answer the following questions. [3]



a) If **X** is a **Th-232** that contains **90** protons, find the atomic number of **Y**
b) Name the special name of the particle **'a.'**
c) State whether the atomic mass of Y increases with the emission of $\mathbf{e^{-1}}$

iii) Observe the displacement time graph of the sound waves given below. Answer the questions that follow. [4]



a) Name the type of vibrations represented by the graph.
b) Find the frequency of the wave.
c) Can it be audible to the human?
d) Give reason for your answer in question (c).

Question 9

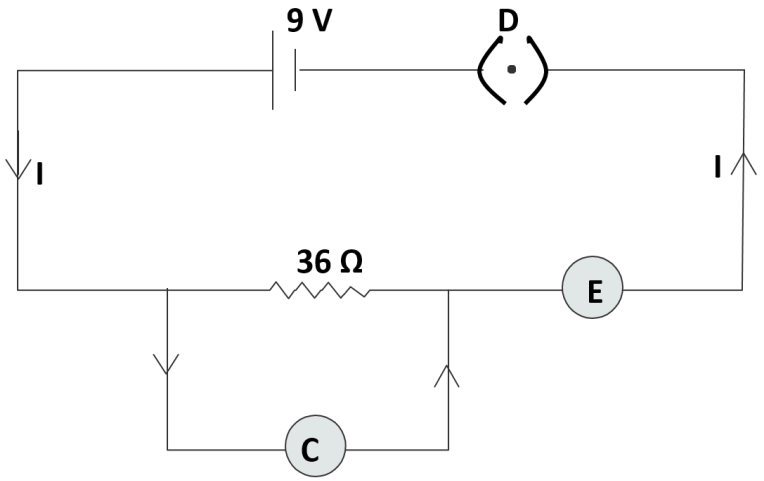
i) (a) A certain amount of heat will warm 1 g of material X by 10°C and 1 g of material Y by 40°C. Which material has a higher specific heat capacity? [3]

(b) Which material, X or Y, would you select to make a calorimeter?

(c) The specific heat capacity of a substance remains the same when it changes its state from solid to liquid. State True or False.

ii) A solid mass of 200 g at 80 °C is dropped in water of mass 50g at 10 °C. Determine the study temperature of the mixture if the specific heat capacities of the solid and water are 0.24 $Jkg^{-1}K^{-1}$ and 4.2 $Jkg^{-1}K^{-1}$ respectively [3]

iii) Consider the circuit diagram given below. Answer the questions that follow. [4]



a) Name the devices C and E in the circuit shown?
b) Find the reading in E.
c) Why is device C connected in parallel to the resistor 36 Ω?

**********All the best **********

Anantha Vidyanikethana
Midterm examination - SEP - 2025
PHYSICS - grade 10 (Key Answers)
(SCIENCE PAPER 1)

Maximum Marks: 80 **Time allowed: Two hours**
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SECTION A (40 Marks)
(Attempt **all** questions from this Section.)

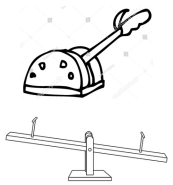
Question 1

i) b) negative [1]
ii) a) 1M kgfm [1]
iii) d) Nuclear → heat → electrical [1]
iv) c) It is convenient to exert effort. [1]
v) a) Refraction of light [1]
vi) a) Image is beyond 24 cm [1]
vii) d) Both b and c [1]
viii) c) The slope is the tangent of the curve [1]
ix) d) Both a and c [1]
x) b) protons = 93, neutrons = 144 [1]
xi) b) Total internal reflection [1]
xii) c) a bangle [1]
xiii) a) 200 V [1]
xiv) c) + 2 D [1]
xv) b) Potassium - 42 [1]

Question 2

i) Complete the following by choosing the correct answers from the brackets: [6]
(a) Class II [1]
(b) convex lens [1]
(c) NmK^{-1} [1]
(d) 20Hz [1]
(e) temperatures [1]
(f) alpha [1]

ii)



Class 2 [1]

Class 1 [1]

(iii) a) ultrasound [1]
c) Frequency more than 20 kHz and travel undeviated long distances. [1]

Question 3

i) a) Total internal reflection. [1]

c) 45° [1]

ii) a) X = I x t = 1.5 x 600 = 900 C [1]

b) V = IR = 1.5 x 9 = 13.5 V [1].

iii) MF = 1.7 x 2500 = 4250 Nm [1+1]

iv) (mct) water = Q

x X 4.2 x 27 = 2730 J [1]

$x = \frac{2730}{4.2\ x\ 13}$ = 50 g [1]

v) (a) Resistance: increases [1]

(b) Resistivity: increases [1]

vi) a) V_1

b) A

vii) (a) P

(b) R

c) 99, 244

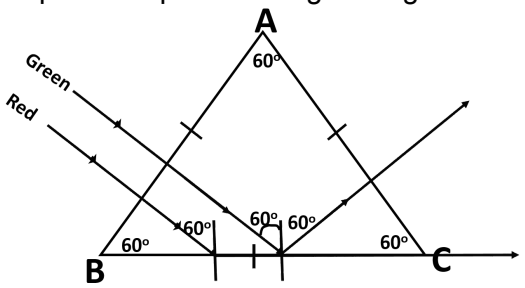
SECTION A (40 Marks)

(Attempt **any 4** questions from this Section.)

Question 4

i) a) 60°

b) Copy the diagram to complete the path of the green light till it emerges from the prism.

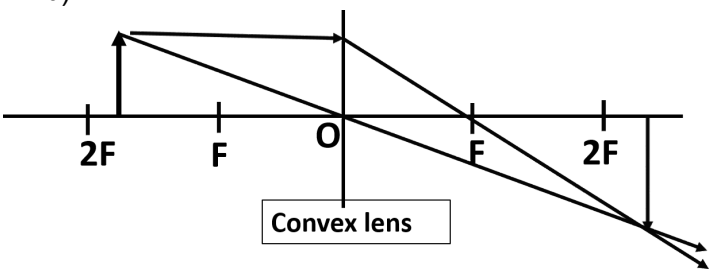


(ii) a) AD = RD/RI = 20/(4/3) = 15 cm [1+1]

b) 30 - 15 = 15 [1]

iii) a) Convex lens [1]

b)



- lens with object placed between F and 2F with proper symbols - [1]
- Ray through optic centre and through F - [1]
- Image [1]

Question 5

i) An object is placed in front of a concave lens at a distance of 45 cm from it. If its image is formed at a distance of 30 cm from the lens, calculate the focal length of the lens.

f = uv/(u-v)

= (-45 x -30)/(-45 + 30) = -90 cm [1+1+1] any other method also accepted

ii) a) BD [1]

b) AC [1]

c) yes [1]

iii) a) 1) Infrared Radiation

2) Infrared Radiation.

3) Ultraviolet (UV) radiation

b) They travel at the speed of light in a vacuum. (implied)

Question 6

i) X and Y are two forces, 2 N and 3 N respectively, 400 cm apart. O is the point 250 cm away from point P on X as shown in the figure. Calculate the sum of moments of X and Y about:

a) Sum of MF at O = (250 x 2 - 150 x 3) = 50 Nm (anticlockwise) [1+1]

b) Sum of MF at P = 400 x 3 = 1200 Nm (Clockwise) [1]

ii) a) Class 3 [1]

b) VR > 1 [1]

c) By increasing the effort arm [1]

iii) a) KE = ½ mv^2 = ½ x 200 x 10 x 10 = 10000 J [1 + 1]

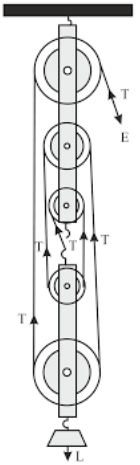
b) h = U/mg = 10000/(200 x 10) = 5 m [1]

c) KE = 10000 - 1000 = 9000 J [1]

Question 7

i) Draw a block and tackle pulley of VR = 5 with a convenient direction of effort. [3]

- Correct pulley blocks with rigid support [1]
- Thread arrangement [1]
- Mentioning load, effort, and tension force [1]



ii) a) Loudness [1]

b) L ∝ A^2

1:9 [1]

d) Resonance [1]

iii) a) d = vt/2 = (1500 x 100)/2 = 75000 m [1+1]

b) t = 2d/v = (2 x 75000)/320 = 468.75 [1+1]

Question 8

i) a) Superconductor, a Conductor that offers zero resistance at very low absolute temperatures. [1+1]

b) No. [1]

ii) a) Atomic number of Y = 90 -4 + 3 = 89 [1]

b) Alpha [1]

c) No. [1]

iii) a) Natural vibrations (free vibrations). [1]

b) F = ¼ = 0.25 Hz (s^{-1}) [1]

c) NO [1]

d) Because human hears with a range of 20 Hz to 20000 Hz. [1]

Question 9

i) (a) X [1]

(b) Y [1]

(c) No. [1]

ii) (mct) water = (mct) copper

300 X 4.2 x (40 - t) = 600 x 0.42 x (t - 10) [1+1]

100 x 10 x (40-t) = 200 x (t-10)

5 (40-t) = t-10

200 + 10 = 210 = 6t

t = 210/6 = 35 ^{o}C [1]

iii) a) C - voltmeter and E = ammeter [1+1]

b) I = V/R = 9/36 = ¼ = 0.25 A. [1]

c) Because in parallel, the Potential difference across the resistor is constant. [1]